📚 Neuromorphic Computing Basics

Brief Overview

This note covers **neuromorphic computing** and was created from the [Architecture All Access: Neuromorphic Computing Part 1]() YouTube video. It captures the video's explanation of how brain-inspired design translates into low-power, high-speed hardware.

Key Points

- Biological neuron models and spike encoding.
- Comparison of brain and conventional computing power.
- Sparse, asynchronous spike communication.
- Foundations for building neuromorphic chips.

Neuromorphic Computing Overview 🧠

> **Neuromorphic computing**: a design approach that mimics the structure and principles of biological brains to achieve high efficiency, speed, and intelligence in digital hardware.

- Inspired by **evolutionary biology**—the brain's billion-year-old solutions to computation.
- Intel's neuromorphic lab, led by **Mike Davies**, focuses on translating cellular-level brain principles into mass-producible chips.

Biological Neural Networks Basics 🧬

> **Neuron**: a cell that processes information via voltage **spikes** (binary events).

> **Spike**: an all-or-none voltage pulse; information is carried by the **timing** of spikes, not their magnitude.

> **Synapse**: the junction where an incoming spike injects ions into the **dendrite** of the receiving neuron, altering its membrane voltage.

> **Soma**: the cell body where dendritic voltage changes are summed; if excitation exceeds a threshold, the soma fires an output spike.

- **Axon**: output channel that branches like a tree to reach many synapses.
- **Excitatory vs. Inhibitory inputs**: excitatory spikes push the soma toward firing; inhibitory spikes pull it away.
- **Connectivity**: each neuron typically links to ~10,000 others, creating dense, recurrent, irregular networks (far beyond standard CMOS layouts).

Key Characteristics

- **Binary spikes** → information encoded in **temporal patterns**.
- Networks are **recurrent** with loops everywhere, not just deep, feed-forward layers.
- Neurons continuously **store and release energy**, analogous to compressing springs.

Energy Efficiency & Speed Comparison 🚀

Feature	Tiny Cockatiel Brain (≈2 g, 50 mW)	Conventional Computer Architecture
Power	~50 mW	Orders of magnitude higher
Weight	2 g	Much heavier hardware
Speed	Enables flight at 20 mph, rapid navigation, tool use, vocalization	Slower for comparable tasks when measured per energy unit
Performance	Outperforms state-of-the-art computers in speed, weight, and power by **orders of magnitude**	Limited by memory bandwidth and serial processing

Spiking Neural Network Example: Visual Object Recognition 🐱

Conventional Pipeline

1. Capture image → array of pixel values.
2. Serialize/ vectorize data.
3. Perform massive **multiply-accumulate (MAC)** operations using large weight matrices at GHz speeds.
4. Repeatedly stream matrices from memory → **memory bandwidth bottleneck**.

Neuromorphic Pipeline

1. **Retina spikes** encode **features** (texture, motion, color) directly, not raw pixels.
2. Spike **timing** reflects feature prominence → creates a **spatiotemporal spike signature**.
3. Signature distributed to **10,000 parallel neurons** operating on a millisecond scale.
4. Each neuron stores a learned temporal pattern; the closer the match, the **faster it spikes**.
5. An **inhibitory feedback network** monitors overall activity and sends a "stop" message once sufficient output spikes appear, **power-gating** the system.
6. Resulting output spikes encode the inference (e.g., "cat detected") with temporal precision.

Comparative Highlights

Aspect	Conventional Deep Learning	Neuromorphic Spiking Network
Data Representation	Raw pixel values → large vectors	Feature spikes → sparse events
Computation	Dense MACs → billions of operations	Sparse event-driven processing
Memory Access	Continuous streaming of large matrices	Minimal, only needed spike data
Latency vs. Energy	Similar latency possible but at **high energy cost**	Low latency **and** low energy due to sparse activity
Scalability	Limited by memory bandwidth and batching	Scales like brain networks (recurrent, sparse)

Advantages of Sparse Asynchronous Communication ⚡

- **Efficiency**: Sparse spikes reduce unnecessary activity, saving power.

- **Low Latency**: Event-driven updates propagate only when changes occur.
- **Sensor Integration**: Sensors can output spikes directly, cutting **output bandwidth**.
- **Robotics Applications**: Immediate response to unpredictable environmental changes.

Computation Paradigm Shift 🌐

> **Emergent computation**: the result of millions of independently executing neurons interacting, causing the network to settle into a new equilibrium that encodes the answer.

- Networks behave as **non-linear, event-driven dynamical systems**.
- **Learning** is also emergent: as the network processes data, it **self-organizes**, adjusting weights toward a more adapted state.
- Neuromorphic chips **search for equilibrium states** rather than following rigid instruction sequences.
- Information propagates via **spikes through sparse recurrent connections**, not through streamed matrices.

Mapping Neuromorphic Concepts to Silicon (Preview) 🔧

- The next step involves translating the described biological principles into **chip architectures** that can be mass-produced.
- Future discussion (Part 2) will cover concrete implementation strategies.